

DispatchFlow — Technical Architecture Blueprint

1. System Overview

DispatchFlow is a multi-tenant logistics and operational workflow platform.

The system is designed to support:

- Multiple organizations
- Multiple branches
- Procurement workflows
- Shipment tracking
- Real-time operational updates
- Mobile-first usage
- File uploads
- Notifications

Architecture priority: 1. Speed of MVP development 2. Reliability 3. Scalability 4. Operational visibility

2. Recommended Architecture

Frontend

- Next.js 15
 - TypeScript
 - Tailwind CSS
 - shadcn/ui
 - React Query
 - Zustand
-

Backend

Recommended MVP Stack

- Supabase
- PostgreSQL
- Supabase Auth
- Supabase Realtime
- Supabase Storage
- Edge Functions

Reason: Fastest route to production.

3. High-Level Architecture

Frontend (Next.js) ↓ API Layer ↓ Supabase Backend ↓ PostgreSQL Database ↓ Storage + Notifications

4. Frontend Structure

Folder Structure

/app /auth /dashboard /requests /shipments /procurement /branches /settings /analytics

/components /ui /forms /tables /timeline /cards /layout

/lib /api /auth /utils /constants

/hooks /services /types /store

5. Main Pages

Authentication

- Login
 - Forgot Password
 - OTP Verification
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Dashboard

Widgets

- Pending Requests
 - Delayed Shipments
 - Deliveries Today
 - Procurement Costs
 - Branch Activity
-

Requests Module

Features

- Create request
 - Edit request
 - Approvals
 - Attachments
 - Request history
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Shipments Module

Features

- Create shipment
 - Shipment timeline
 - Status updates
 - Transit tracking
 - Delivery confirmation
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Procurement Module

Features

- Supplier tracking
 - Invoice uploads
 - Cost tracking
 - Purchase records
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Analytics Module

Features

- Delivery metrics
 - Branch performance
 - Delay analysis
 - Operational reports
-

6. Database Design

USERS TABLE

Field	Type
id	uuid
full_name	text
email	text
phone	text
role	text
company_id	uuid
branch_id	uuid
created_at	timestamp

COMPANIES TABLE

Field	Type
id	uuid
name	text
logo_url	text
created_at	timestamp

BRANCHES TABLE

Field	Type
id	uuid
company_id	uuid
city	text
province	text

Field	Type
address	text
created_at	timestamp

REQUESTS TABLE

Field	Type
id	uuid
company_id	uuid
branch_id	uuid
requester_id	uuid
title	text
priority	text
status	text
notes	text
created_at	timestamp

REQUEST_ITEMS TABLE

Field	Type
id	uuid
request_id	uuid
item_name	text
quantity	integer
unit_price	numeric
total_price	numeric

PROCUREMENTS TABLE

Field	Type
id	uuid
request_id	uuid
supplier_name	text
purchase_cost	numeric
invoice_url	text
purchased_by	uuid
created_at	timestamp

SHIPMENTS TABLE

Field	Type
id	uuid
request_id	uuid
tracking_code	text
transport_company	text
origin	text
destination	text
status	text
estimated_delivery	timestamp
created_at	timestamp

SHIPMENT_UPDATES TABLE

Field	Type
id	uuid
shipment_id	uuid

Field	Type
location	text
status	text
notes	text
updated_by	uuid
created_at	timestamp

ATTACHMENTS TABLE

Field	Type
id	uuid
request_id	uuid
shipment_id	uuid
file_url	text
uploaded_by	uuid
created_at	timestamp

ACTIVITY_LOGS TABLE

Field	Type
id	uuid
user_id	uuid
action	text
entity_type	text
entity_id	uuid
metadata	jsonb
created_at	timestamp

7. Authentication & Permissions

Authentication

Use Supabase Auth.

Methods

- Email/password
 - OTP
 - Magic links
-

Authorization

Implement RBAC.

Roles

- employee
 - branch_manager
 - procurement_officer
 - dispatch_officer
 - regional_manager
 - super_admin
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8. Status Architecture

REQUEST STATUSES

- pending_approval
 - approved
 - rejected
 - purchasing
 - purchased
 - dispatch_ready
 - shipped
 - delivered
 - completed
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SHIPMENT STATUSES

- preparing
 - dispatched
 - in_transit
 - arrived_transit_point
 - delayed
 - delivered
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9. Real-Time Updates

Use Supabase Realtime.

Live Features

- Shipment updates
 - Request approvals
 - Dashboard refresh
 - Notifications
-

10. Notification Architecture

WhatsApp

Use:

- WhatsApp Cloud API or
 - Twilio
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Email

Use:

- Resend or
 - SendGrid
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SMS

Use:

- Twilio
 - Africa's Talking
-

11. File Upload Architecture

Use Supabase Storage.

Upload Types

- Images
 - PDFs
 - Invoices
 - Delivery signatures
 - Waybills
-

12. Security Requirements

Required

- Row-level security
 - File validation
 - Rate limiting
 - JWT authentication
 - Permission middleware
 - Audit logs
-

13. API Design

REQUEST ENDPOINTS

POST /requests GET /requests GET /requests/:id PATCH /requests/:id DELETE /requests/:id

SHIPMENT ENDPOINTS

POST /shipments GET /shipments GET /shipments/:id PATCH /shipments/:id

UPDATE ENDPOINTS

POST /shipment-updates GET /shipment-updates/:shipmentId

14. UI Components

Core Components

- Data tables
 - Timeline component
 - File uploader
 - Status badge
 - Activity feed
 - Notification center
 - Dashboard cards
 - Mobile navigation
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15. Mobile UX Requirements

Mandatory

- Bottom navigation
 - Offline capability
 - Camera integration
 - Fast forms
 - Low bandwidth optimization
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16. Deployment Architecture

Frontend

Deploy on:

- Vercel
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Backend

Deploy on:

- Supabase Cloud
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Domain Setup

- app.dispatchflow.africa
 - api.dispatchflow.africa
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17. Monitoring & Logging

Monitoring

Use:

- Sentry
 - LogRocket
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Analytics

Use:

- PostHog
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18. Scaling Roadmap

PHASE 1

MVP

PHASE 2

Multi-company support

PHASE 3

Advanced analytics

PHASE 4

Fleet tracking

PHASE 5

Inventory management

PHASE 6

ERP integrations

19. Future AI Features

Potential AI Modules

- Delivery delay prediction
 - Smart procurement recommendations
 - Fraud detection
 - Operational optimization
 - Auto-generated reports
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20. Development Roadmap

WEEK 1

Database + Auth

WEEK 2

Request module

WEEK 3

Shipment module

WEEK 4

Timeline + notifications

WEEK 5

Dashboard + analytics

WEEK 6

Testing + deployment

21. Production Priorities

Priority order:

1. Reliability
2. Operational visibility
3. Mobile usability
4. Accountability
5. Real-time updates
6. Speed
7. UI polish

Do not optimize aesthetics before operational clarity.